Brandon Yifan Yang

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Education

- **University of Pennsylvania** Philadelphia, PA
  M.S.E. in Robotics; GPA: 3.7/4.0 *May 2027*
- **University of Virginia** Charlottesville, VA
  B.S. in Computer Science; Highest Distinction; GPA: 3.9/4.0 *May 2025*
- *Selected coursework*: Learning in Robotics, Reinforcement Learning, Probabilistic Machine Learning, GPU Programming & Architecture, Physical Intelligence, Natural Language Processing
- *Teaching*: Lead TA for Computer Systems Organization (2x), Theory of Computation, and Machine Learning (2x)

Experience

Research Intern: Agentic Autonomous Vehicles Mountain View, CA
Honda Research Institute USA, Inc. *September 2026 - December 2026*

- Researching gradient-free, self-improving autonomous physical AI systems with frontier model agents.

Research Lead San Francisco, CA
AfterQuery (YC W25) *May 2026 - Aug 2026*

- Led development of a custom-designed UMI system; built a new hardware-integration library and end-to-end software for demonstration collection, motion tracking, SLAM, data filtering and storage, and robot imitation policy training and evaluation with dual YAM robot arms.
- Led a 6-person research team building a full-stack AI-agent search benchmark platform; collected tasks from external experts, built custom quality-control workflows, and automated evaluations across frontier agent harnesses and models.

Embodied Intelligence Research Intern Beijing, China
Spirit AI *May 2025 - Aug 2025*

- Trained and evaluated VLA policy variants in PyTorch across simulation and real dual-arm robot experiments, comparing contrastive objectives, observation-noise curricula, reduced guidance inputs, and mixture-of-experts architectures.
- Designed and implemented a DAgger data-collection pipeline for VLA models, including operator-takeover logic and corrective-demonstration workflows used for post-training and evaluation.
- Built a reproducible Docker/Slurm pipeline for VLA simulation, training, and evaluation workflows; open-sourced code and datasets on GitHub.

Research Experience

Perception, Action, & Learning Lab, University of Pennsylvania Philadelphia, PA
Advisor: Dinesh Jayaraman *Aug 2025 - Present*

- Implemented demonstration retrieval for $\pi_{0.5}$ VLA policies, integrating robot data retrieval into training and test-time inference for tabletop manipulation experiments.
- Engineered data transformation, training, and evaluation pipelines for retrieval-based VLA variants using JAX, PyTorch, Hugging Face, Docker, Slurm, and Ray.
- Designed hidden-activation collection and steering infrastructure for diffusion-policy, flow-matching, and autoregressive VLA models across MetaWorld, LIBERO, RoboCasa, and real Franka hardware experiments. (*ICML 2026 Workshop; ICLR 2027 under review*)

Learning and Interactive Robotics, University of Virginia Charlottesville, VA
Advisor: Yen-Ling Kuo *Aug 2024 - May 2025*

- **SkillVLA**: Developed a skill-conditioned VLA model for language-conditioned manipulation, using subgoal instructions and a VQ-VAE latent skill library to structure action generation.

- Presented SkillVLA as an oral talk at the 2024 UVA LLM Workshop (slides), earning the Audience Choice Award (top 3 of 28 presentations).
- **Contrastive Learning for Robot Manipulation**: Modified simulation environments and trained PyTorch visuomotor policies for CLASS to evaluate action-sequence-supervised visual embeddings under camera-pose and object-appearance shifts. (CoRL 2025)

Collaborative Robotics Lab, University of Virginia Charlottesville, VA
Advisor: Tariq Iqbal *May 2022 - May 2024*

- **Grounded Location for Object Manipulation (GLOMA)** [code]: Led a team of 3 in developing a language-grounded, bounding-box-guided image-editing pipeline for object relocation in robotic manipulation; collected and annotated a dataset for fine-tuning pre-trained language and vision models.
- **Centralized Multi-Agent RL for Collaborative Tasks** [code]: Implemented offline and online centralized MARL training for long-horizon robotic bolt-screwing tasks in Isaac Gym.
- Designed custom multi-agent reward functions for task completion and collaboration, improving simulated task success by 20% over the project baseline.

Publications

Amish Sethi, Peter Jiashu Shen, Aiden Ryou, **Brandon Y. Yang**, Spencer Mateega, and Sam Jung. "CrossSearch: Benchmarking Multimodal Agents on Tasks Requiring Both Local Files and Live Web Evidence." Submitted to *ACM TheWebConf 2027, Conference Long Papers*.

Govind Velamoor, Adrian Lin, Sam Jung, Spencer Mateega, **Brandon Y. Yang**†, and Amish Sethi†. "UMI-Live for In-The-Wild Bimanual Robotics Data Collection." *CoRL 2026 UMI Arena Workshop*.

Jiani Huang*, Amish Sethi*, **Brandon Y. Yang***, Yuchen Zheng, Christopher Watson, Jianing Qian, Mayur Naik, and Dinesh Jayaraman. "Retrieval-Augmented Vision-Language-Action Model." *CoRL 2026 SimBench2Real Workshop*.

Miranda Muqing Miao*, Subin Kim*, **Brandon Y. Yang***, Dinesh Jayaraman, and Lyle Ungar. "Contrastive Conceptor Activation Steering (COAST): Unlocking Vision-Language-Action Models through Hidden States." *ICML 2026 Mechanistic Interpretability Workshop*; under review at *ICLR 2027*. [Workshop][arXiv]

Sung-Wook Lee, Xuhui Kang, **Brandon Y. Yang**, and Yen-Ling Kuo. "CLASS: Contrastive Learning via Action Sequence Supervision for Robot Manipulation." Accepted to *CoRL 2025*. [Website][arXiv][PDF]

*Equal contribution among starred authors. †Equal advising.

Software Projects

openpi-eval [GitHub, Dataset, X Post]: Built single-repo OpenPI evaluation infrastructure for a NeurIPS submission, supporting reproducible $\pi_{0.5}$ rollouts across MetaWorld, LIBERO, LIBERO-Pro, RoboCasa, and RoboLab; server-side batching improved policy-server throughput by up to 3.4x in a 64-client LIBERO benchmark.

agentic-robots [GitHub, X Post]: Built an agent-driven control framework for dual YAM robot arms, connecting Codex to camera observations and robot actions through MCP; supports closed-loop visual feedback, action correction, and rollout recording.

openpi-cuda [GitHub]: Developed custom CUDA kernels with C++/Python bindings for $\pi_{0.5}$ VLA inference, reducing latency by 18.5ms versus the PyTorch baseline.

notie-markdown [Website]: Built a React/TypeScript markdown renderer with equation previewing, graphing, and code execution; shipped with a Python/Flask backend on Vercel and Heroku.

Skills

Programming: Python, C++, CUDA, TypeScript/JavaScript, Java

ML & Robotics: PyTorch, JAX, Hugging Face, Ray, Slurm, robot learning, VLA models

Web & Infrastructure: React, FastAPI, Flask, Docker, AWS EC2, Vercel, `uv`, `ruff`, `ty`

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